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Giant Negative Electrocaloric Effect in (Pb,La)(Zr,Sn,Ti)O₃ Antiferroelectrics near Room Temperature

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ABSTRACT: $(\text{Pb}_{0.97}\text{La}_{0.02})(\text{Zr}_x\text{Sn}_{0.94-x}\text{Ti}_{0.06})\text{O}_3$ (PLZST) antiferroelectric ceramics with $x = 0.75\text{--}0.90$ have been fabricated and found to be a novel electrocaloric material system with a giant negative electrocaloric effect ($\Delta T = -11.5$ K) and a large electrocaloric strength ($|\Delta T/\Delta E| = 0.105$ K cm kV^{-1}) near room temperature. Additionally, the PLZST antiferroelectric ceramic also exhibits a large positive electrocaloric effect around the Curie temperature. The giant negative effect and the coexistence of both positive and negative electrocaloric effects in one material indicate a promising possibility to develop mid- to large-scale solid-state cooling devices with high efficiency.

1. INTRODUCTION

Electrocaloric (EC) refrigeration has attracted tremendous attention recently and emerged as a viable means to realize primary solid-state cooling technology because of the substantial increase in efficiency of current cooling devices and the concern of global warming.^{1–5} Compared to conventional vapor-compression refrigeration, cooling technologies based on the electrocaloric effect (ECE) do not require clumsy configuration and are more efficient.^{3,6} ECE, the reversible temperature change (ΔT) and corresponding entropy change (ΔS) of a dielectric material under the removal or application of an electric field, has been studied for many decades since the 1930s.⁷ However, for a long time, the ECE was eventually abandoned due to the small temperature changes ($\Delta T < 2$ K) obtained in bulk single crystals or ceramics.⁸ Only recently, a giant ECE with a ΔT value of 12 K under an electric field change (ΔE) of 480 kV cm^{-1} has been pronounced in the $\text{PbZr}_{0.95}\text{Ti}_{0.05}\text{O}_3$ (PZT95/05) thin film near its Curie temperature (T_C) of 222 °C, which fuels enormous interest in this field.⁹ After that, a similarly large ΔT (12.6 K at 2090 kV cm^{-1}) has been achieved in the poly(vinylidene fluoride-trifluoroethylene) [P(VDF-TrFE)] ferroelectric (FE) copolymer above the T_C of 70 °C.¹⁰

Subsequently, researchers have made great efforts to enhance the ECE. For example, Peng et al. have reported a large ECE ($\Delta T = 45.3$ K at 598 kV cm⁻¹) in Pb_{0.8}Ba_{0.2}ZrO₃ (PBZ) relaxor ferroelectric thin film¹¹ and Zhao et al. have also achieved a larger ECE ($\Delta T = 53.8$ K at 900 kV cm⁻¹) in Pb_{0.97}La_{0.02}(Zr_{0.75}Sn_{0.18}Ti_{0.07})O₃ (PLZST2/75/07) antiferroelectric (AFE) thin film.¹² Those outstanding ECEs offer the potential for highly efficient and environmentally friendly solid-state on-chip cooling devices.

Interestingly, recent experimental and theoretical studies have demonstrated that two types of ECEs exist in relaxor FE and AFE materials: conventional (positive) ($\Delta T > 0$) and anomalous (negative) effect ($\Delta T < 0$).^{13–16} Compared to the cooling device using positive ECE only, the cooling efficiency of a device by combining negative and positive ECEs could be much improved.¹⁷ Ponomareva et al. have proposed that ferroelectrics with multiple phase transitions may exhibit the coexistence of both negative and positive ECE in one material.¹⁸ Excitingly, a giant negative ECE about -5.0 K under a field of 308 kV cm⁻¹ has been found in (Pb_{0.97}La_{0.02})(Zr_{0.95}Ti_{0.05})O₃ (PLZT2/95/05) AFE thin film.¹⁹ Very recently, Park et al. have also reported a giant negative ECE ($\Delta T = -10.8$ K at 3260 kV cm⁻¹) in Hf_{0.5}Zr_{0.5}O₂ (HZO50/50) FE thin film.²⁰ The giant negative ECEs in thin films will propose novel combinations with positive ECE to further enhance the electrocaloric efficiency. Therefore, giant negative ECE can be considered as another alternative solution for next-generation solid-state cooling processes.

Although giant positive or negative EC effects have been reported in AFE or FE thin films under large operating electric fields, their cooling efficiencies are relatively lower than those of bulk EC materials under the same large electric fields due to the small volume, and so EC strength $|\Delta T/\Delta E|$ tends to be disproportionately small.^{21–24} In fact, the quest for higher ΔT in bulk EC materials continues because high ΔT is imminently required for the midium- and large-scale solid state cooling devices. Unfortunately, to date the reported ΔT values of the bulk ceramics and single crystals are relatively small ($|\Delta T| < 5$ K).^{25–32} This is

understandable because the entropy or polarization change near the phase transition becomes smaller once a higher field is employed.^{3,33-35} Therefore, finding new EC materials with large $|\Delta T|$ and $|\Delta T/\Delta E|$ values, and understanding the underlying mechanism are of paramount importance in solid state cooling technology. Additionally, an efficient EC response near room temperature is urgently needed to develop practical cooling devices. Here, we report a giant negative ECE ($\Delta T = -11.5$ K) and a significantly enhanced EC strength ($|\Delta T/\Delta E| = 0.105$ K cm kV⁻¹) in (Pb_{0.97}La_{0.02})(Zr_xSn_{0.94-x}Ti_{0.06})O₃ ($x = 0.75-0.90$) bulk antiferroelectric ceramics near room temperature. A physical explanation based on the Ginzburg-Landau-Devonshire phenomenology for this extraordinary ECE is also proposed.

2. EXPERIMENTAL PROCEDURE

The (Pb_{0.97}La_{0.02})(Zr_xSn_{0.94-x}Ti_{0.06})O₃ ceramics with $x = 0.75, 0.80, 0.85$, and 0.90 (denoted as PLZST2/75/06, PLZST2/80/06, PLZST2/85/06, and PLZST2/90/06, respectively) were fabricated using solid-state reaction method with high purity (>99.99%) raw materials of La₂O₃, PbO, SnO₂, ZrO₂, and TiO₂.^{36,37} The calcination was carried out at 900–950 °C for 2 h, then the sintering was performed at 1250–1350 °C in air for 4 h. For electrical property characterizations, Au electrodes were sputtered on both sides of the ceramic plate (2 mm × 2 mm × 0.4 mm) using an ion sputtering instrument (ETD-3000, COXEM Co., Ltd). The basic structural characterizations of X-ray diffractometer (XRD, Brüker D8 ADVANCE) with Cu K α radiation at $2\theta = 10^\circ-70^\circ$ and scanning electron microscopy (SEM, Quanta 200 FEG) images of the as-prepared ceramics were carried out. Raman spectra at various temperatures were performed using a micro-Raman spectrometer (HR-800, Horiba). Thermal characteristics of all the ceramics were measured using a commercial differential scanning calorimeter (DSC, Q2000). Temperature-dependent dielectric properties were measured by an Agilent 4294A impedance analyzer (Agilent Inc., Bayan). Polarization vs electric field hysteresis ($P-E$) loops were measured at 10 Hz using a TF 2000E ferroelectric tester (aix

ACCT Co., Aachen). The leakage current measurements were recorded using a Radiant ferroelectric system (Precision Premier II, Radiant Technologies Inc., Albuquerque).

3. RESULTS AND DISCUSSION

Room temperature XRD on the PLZST ceramics was carried out to study the phase structure. As shown in Figure 1a, the as-prepared PLZST ceramics have a pure perovskite structure.¹² In order to determine the structure of the ceramics, {200} and {211} reflection peaks were performed at a step size of 2.0, which are displayed in Figures 1a,c. It can be seen that the ceramics with $x \geq 0.80$ display the (002)/(200) splitting ($2\theta \approx 44^\circ$) and separated (211)/(112) peaks ($2\theta \approx 54.5^\circ$), indicating an antiferroelectric orthorhombic (AFE_O) symmetry, as observed in many PbZrO₃-based ceramics.³⁷ For PLZST ceramics with $x = 0.75$, a small splitting peaks were observed, revealing an antiferroelectric tetragonal (AFE_T) symmetry.¹² Our results suggest that the patterns of PLZST ceramics with $x \geq 0.80$ show an AFE_O phase, while the PLZST ceramics with $x = 0.75$ give an AFE_T phase. The average grain size of the PLZST ceramics is about 6–7 μm , as shown in Figure S1 in Supporting Information.

Figure 2a plots the temperature dependence of dielectric permittivity (ϵ) at different frequencies for the PLZST2/80/06 ceramics. Two anomalies in dielectric permittivity are observed, indicating the phase transition temperatures. PLZST2/80/06 is in AFE_O phase at room temperature, which transforms to AFE_T phase at 70 °C and then to cubic paraelectric phase (PE_C) at the Curie temperature T_C (≈ 202 °C) on heating. The AFE_O→AFE_T→PE_C phase transition sequence could be confirmed by the temperature-dependent Raman spectra (Figure S3) and the heat capacity C_p (Figure 2b). A sharp peak in C_p – T curve, corresponding to the phase transition between AFE_T and PE_C phases, is found around T_C . However, a slight increase in the heat capacity is measured at the AFE_O to AFE_T phase transition temperature (T_{O-T}). The AFE_O→AFE_T→PE_C phase transition sequence can also be found in the

PLZST2/85/06 and PLZST2/90/06 ceramics, while the PLZST2/75/06 ceramics undergo AFE_T→PE_C phase transition during heating (Figure S2 and Figure S3).

Temperature-dependent polarization $P(T)$ of the PLZST2/80/06 ceramics under various applied electric fields were recorded at 10 Hz (Figure 2c). Interestingly, the polarization change with temperature is nonmonotonic with a maximum value at T_M . The polarization increases first when $T < T_M$ and then decreases with further heating (typical behavior of ferroelectric materials under high field). The T_M displays a linear shift toward low temperatures with a large rate $dT_M/dE = -0.46 \text{ K cm kV}^{-1}$ (see Figure 2d). Emergence of the ferroelectric behavior can also be verified by the temperature-dependent AFE to FE switching field ($E_{\text{AFE-FE}}$) obtained from the P - E hysteresis loops (see Figure S4), as shown in the inset of Figure 2d. Similar $P(T)$ behaviors are found in the PLZST2/75/06, PLZST2/85/06, PLZST2/90/06 ceramics (Figure S5) and reported in the PLZT2/95/05 thin films.¹⁹

To investigate the electrocaloric effect, the Maxwell relationship is used:³⁸

$$\left(\frac{\partial S}{\partial E}\right)_T = \left(\frac{\partial P}{\partial T}\right)_E \quad (1)$$

where S is the entropy, E is the applied electric field, T is the temperature, and P is the macroscopic polarization, respectively. Reversible changes in temperature ΔT and entropy ΔS for an electrocaloric material in response to a field change $\Delta E = E_2 - E_1$ can be expressed as:

1,39-45

$$\Delta T = -\frac{1}{C_p \rho} \int_{E_1}^{E_2} T \left(\frac{\partial P}{\partial T}\right)_E dE \quad (2)$$

$$\Delta S = -\frac{1}{\rho} \int_{E_1}^{E_2} \left(\frac{\partial P}{\partial T}\right)_E dE \quad (3)$$

where C_p is the heat capacity and ρ is the density of the material. Reversible changes in temperature ΔT and entropy ΔS can be calculated by numerically solving eqs 2 and 3 using MATHEMATICA (and using the $\rho = 7.9 \text{ g cm}^{-3}$ measured by the Archimedes method and the

heat capacity C_p taken from the experimental data as shown in Figure 2b). Here, for our case, the AFE state ($E_1 = 0$) is adopted instead of deliberately employing a high field E_1 (larger than $E_{\text{AFE-FE}}$) to achieve the FE state. Figure 3a shows the temperature changes in ΔT for the PLZST2/80/06 ceramics as a function of temperature. As the electric field E_2 increases from 20 kV cm⁻¹ to 110 kV cm⁻¹, the negative ΔT peaks are found at different temperatures. Interestingly, the negative ΔT peak shifts remarkably toward lower temperatures, which is accompanied by a significant increase in the magnitude of $|\Delta T|$. Giant negative ECE with a value of $\Delta T = -11.5$ K ($\Delta S = -12.5$ J kg⁻¹ K⁻¹) under a modest field of $\Delta E = E_2 - E_1 = 110$ kV cm⁻¹ and a large electrocaloric strength $|\Delta T/\Delta E| = 0.105$ K cm kV⁻¹ are achieved at 30 °C. Table 1 compares the EC properties of the PLZST2/80/06 ceramics and corresponding conditions used in this work to those in the literature.^{9-13,16,19,20,22-33} The values of the negative ΔT and $|\Delta T/\Delta E|$ of the PLZST2/80/06 ceramics are the largest among the pure electrocaloric materials known up to date. Notably, the negative ΔT of the PLZST2/75/06, PLZST2/85/06, and PLZST2/90/06 ceramics are -5.6 K, -12.8 K, and -14.1 K under a field of 110 kV cm⁻¹ at 30 °C, 40 °C, and 50 °C, respectively, as displayed in Figures S5 and S7. Vats et al. achieved the largest negative ECE value of -52.2 K in a PbZr_{0.53}Ti_{0.47}O₃/CoFe₂O₄ (PZT/CFO) ferroelectric/ferromagnetic multilayer nanostructure made of three layers of PZT and CFO films due to the interplay of the ECE from two functional films.⁶ Table 1 focuses on pure materials and the ECE of multilayer capacitors based on the PLZST bulk ceramics or thin films would be an interesting research topic in future cooling devices.

As discussed earlier, the PLZST ceramics satisfy most of the essential features required for the midium- to large-scale solid-state refrigeration. For instance, the giant negative ECE ($|\Delta T| > 5$ K) near room temperature and the electric field tunable negative EC response at a wide temperature range are obtained in the PLZST bulk ceramics. Similar electric field tunable EC response has also been found in the Pb_{0.97}La_{0.02}(Zr_{0.75}Sn_{0.16}Ti_{0.09})O₃ (PLZST2/75/09) thin film.⁴⁰ Moreover, the EC response goes from negative to positive once

the temperature is higher than T_M (see Figure 2 and Figure S5). The maximum positive ECE and EC strength values of the PLZST2/80/06 ceramics around T_C are 4.8 K and 0.044 K cm kV^{-1} (see Table S1), respectively. The coexistence of negative and positive ECEs in one material could provide a new way to enhance the cooling efficiency of solid-state refrigerator.^{17,18,46} The above mentioned main features are illuminated more clearly in Figure 3b. In contrast to the high operating temperatures required for the positive ECE, the negative effect can be used in a wide temperature range even near room temperature. Compared with the negative effect reported in the bulk PbZrO_3 (PZO) ceramics³⁰ and $\text{Pb}_{0.99}\text{Nb}_{0.02}(\text{Zr}_{0.85}\text{Sn}_{0.13}\text{Ti}_{0.02})_{0.98}\text{O}_3$ (PNZST13/2/2) ceramics³² (Figure 3b and Table 1), our results show a significant enhancement in both ECE and EC strength (see Figure 3b). More importantly, the obtained negative ECE is larger than that reported in many thin films, such as Eu-doped PbZrO_3 ,³³ PLZT2/95/05¹⁹ and HZO50/50²⁰. The magnitude $|\Delta T|$ of the achieved negative ECE ($-11.5 \text{ K}@110 \text{ kV cm}^{-1}$ at 30°C) is comparable to those obtained from the positive ECE in the P(VDF-TrFE)55/45 ferroelectric copolymer¹⁰ ($12.6 \text{ K}@2090 \text{ kV cm}^{-1}$ at 80°C) and PZT95/05 film⁹ ($12 \text{ K}@480 \text{ kV cm}^{-1}$ at 222°C), and is 4-6 times larger than those obtained in bulk $\text{Pb}_{0.99}\text{Nb}_{0.02}(\text{Zr}_{0.75}\text{Sn}_{0.20}\text{Ti}_{0.05})_{0.98}\text{O}_3$ (PNZST20/5/2) ceramics²⁶ ($2.6 \text{ K}@30 \text{ kV cm}^{-1}$ at 30°C) and BaTiO_3 (BTO) single crystals²² ($1.6 \text{ K}@10 \text{ kV cm}^{-1}$ at 139°C). Our EC strength $|\Delta T/\Delta E|$ here is $0.105 \text{ K cm kV}^{-1}$, which is much larger than the previous results in thin films (see Figure 3b and Table 1). This demonstrates a significant enhancement in the electrocaloric properties reported in EC materials.⁴³⁻⁴⁶ A giant negative magnetocaloric effect ($-6.2 \text{ K}@$ a moderate field of 2 T at 317 K) has been achieved in Heusler-type Ni-Mn-In-(Co) alloys, which is also driven by structural phase transitions.⁴⁷ Therefore, structural phase transitions may be useful for improving caloric effects. Nevertheless, further development of solid-state cooling devices based on the PLZST AFE ceramics are beyond the scope of this investigation, we suggest that the direct measurement should be performed to confirm our indirect measurement results here.

Leakage currents as a function of time (t) (see Figure S8 in Supporting Information) were investigated under the maximum applied electric field (110 kV cm^{-1}) at different temperatures. The measured transients persist more than 1000 ms, and no breakdown occurs even after repetitive testing. Obviously, the leakage currents show a strong initial-time dependence due to the Curie–von Schweidler type dielectric relaxation.⁴⁸ The obtained steady-state small leakage current ($< 1 \mu\text{A}$) can yield negligible Joule heating ($< 10^{-3} \text{ K}$)⁹ and does not affect the P – E results because hundreds of microampere currents are required to switch the polarizations measured at 10 Hz.

The rich physics of antiferroelectricity and ferroelectricity in PLZST system have been widely studied,^{37,49} but some key aspects of the negative ECE in the antiferroelectric PLZST system still need to be addressed. Up to now, the investigation of negative ECE is still in its infancy, and the underlying mechanism of the negative ECE is unclear. Some theoretical insights and experimental studies have been carried out to explain the origin of negative ECE in FE and AFE materials. For example, Ponomareva et al. predicted that both positive and negative ECEs can be simultaneously obtained in one material using first-principles-based simulations on $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{TiO}_3$ alloys.¹⁸ After that, the coexistence of both ECEs has been proposed by lattice-based Monte-Carlo simulations in the presence of defects and phase-field modeling in BaTiO_3 .⁵⁰ Recently, Geng et al. suggested a possible physical mechanism of the negative ECE in AFEs with the consideration of microscopic local polarization.¹⁹ One interesting point (not receiving much attention to date) is that the negative ECE materials undergo an AFE–FE phase transformation upon the application of an electric field. Very recently, we reported a double weak negative EC response (-0.054 K at $25 \text{ }^\circ\text{C}$ and -0.14 K at $125 \text{ }^\circ\text{C}$) in the PLZST single crystal with chemical composition near the AFE/FE phase boundary, which can be regarded as a result of electric field-induced polarization noncollinearity.¹⁶ The giant negative ECE obtained in this study, however, results from a macroscopic effect, where most induced polarization of PLZST ceramics are aligned in the

direction of the external electric field. Therefore, the negative ECE here cannot be fully understood by the mechanisms proposed in previous investigations, thus a very disparate model is required. Based on the Kittel's theory,⁵¹ the Ginzburg-Landau-Devonshire (GLD) phenomenology based on the simple two-sublattice model is introduced to explain the giant negative ECE in the present PLZST ceramics.

We employed the GLD phenomenological theory to explain the field induced phase transitions in PLZST based on the Kittel model. According to Kittel's theory, this scenario can be described by the simple two-sublattice model, i.e., P_1 and P_2 (see Figure S9), and the free energy (F) of the PLZST system can be written as:³⁰

$$F = F_0 + a(P_1^2 + P_2^2) + bP_1P_2 + c(P_1^4 + P_2^4) + d(P_1^6 + P_2^6) - (P_1 + P_2)E \quad (4)$$

where a , b , c and d are the Landau coefficients. The equilibrium values of P_1 and P_2 can be obtained by numerically solving the thermodynamic conditions, $\partial F/\partial P_1 = \partial F/\partial P_2 = 0$, which lead to the relations:

$$2aP_1 + 4cP_1^3 + 6dP_1^5 + bP_2 - E = 0 \quad (5a)$$

$$2aP_2 + 4cP_2^3 + 6dP_2^5 + bP_1 - E = 0 \quad (5b)$$

The FE polarization order parameter (P) and AFE structural order parameter (q) can be given by:

$$P = (P_1 + P_2)/\sqrt{2}, \quad q = (P_1 - P_2)/\sqrt{2} \quad (6)$$

Thus, the free energy can be described in a two-dimensional (2D) phase space of P and q ,

$$F = a(P^2 + q^2) + \frac{b}{2}(P^2 - q^2) + \frac{c}{2}(P^4 + q^4 + 6P^2q^2) + \frac{d}{4}(P^6 + q^6 + 15P^4q^2 + 15P^2q^4) - \sqrt{2}PE \quad (7)$$

Among all the coefficients, a is assumed to be temperature dependent, i.e., $a = a_0(T - T_0)$, where T_0 is related to the Curie temperature, i.e., $T_0 = (66x + 150)/T_c$. In order to capture the main feature of the field induced phase transitions, we assume values to coefficients in the GLD free energy, $a_0 = 2x - 0.6$, $b = 0.2$, $c = 1/(40x - 26)$, $d = 1/3$. GLD phenomenological

theory calculated results reveal that the magnitude of applied electric field is a crucial ingredient to induce a metastable polar FE phase that can stay in a well-defined temperature range. In the case of PLZST2/80/06 ceramics with AFE_O phase at low temperatures, the free energy contour has four minima (two global minima and two local minima), as shown in Figure 4a. The global minimum is the nonpolar AFE phase, where $q \neq 0$ and $P = 0$, while the local minimum corresponds to the induced metastable polar FE phase with $q = 0$ and $P \neq 0$. When temperature is low, the energy barrier between the local minimum and the global minimum is very large. Thus, the system will be stuck in a deep energy well and an electric field driven phase transition trajectory going from B to A is practically forbidden (Figure 4d). As the temperature is raised near T_M (Figure 2), there is enough kinetic energy for the system to hop over the smaller energy barrier (see Figures 4b,d) and the electric field induced trajectory going from B to A is feasible under the same applied electric field, which is accompanied by a generation of a large induced polarization. Relevant structures of the PLZST system containing AFE_O, AFE_T, and induced metastable FE_R phases are schematically represented in Figure 4c. There is a significant improvement in polarization with the temperature increasing ($\partial P/\partial T > 0$), as plotted in Figure 4e, leading to a negative ECE according to eq 2. The numerical results for adiabatic temperature change ATC (ΔT^*) (see Figure 4f) also confirm the negative EC response. Moreover, the calculated ΔT^* shifts toward lower temperatures with increasing the electric field, which is in good agreement with the experimental results (Figure 3a). Note that the shallow minima of the metastable ferroelectric state would result in a rapid transform back to the AFE phase upon the removal of the electric field, suggesting a typical characteristic of the AFE phase.⁵² In addition, the polarization decreases significantly ($\partial P/\partial T < 0$) when temperature is increased up to T_c , where the free energy difference between the nonpolar and polar states becomes very small, leading to the generation of a positive ECE peak (see Figure 3a and Figure 4f). A clearer physical picture of the field induced phase transition can be found in the Figure 5. Figure 5a gives the free energy

contours for the PLZST ceramics at different temperatures. At low temperatures, the antiferroelectric phase is persistently stable because the huge energy barrier between the local minimum and the global minimum. Even though a relatively high electric field is applied to the PLZST system, there is not enough kinetic energy to hop over the energy barrier in the minimum free energy paths (see Figure 5b). Thus, at low temperature, the electric field induced antiferroelectric to ferroelectric phase transition is forbidden, which generates a small polarization response. With increasing temperature and thermal fluctuations, there is enough kinetic energy to traverse the energy barrier, which results in a huge polarization response. The energy barrier becomes deeper with the increasing of composition (x) (see Figure 5b), i.e., higher temperature is needed to trigger the electric field induced antiferroelectric to ferroelectric phase transition under an electric field of the same magnitude. Thus, the negative ECE would be obtained at higher temperature with increasing the composition (see Figure 2 and Figure S6).

It should be emphasized, however, that the triple well energy landscape (b) is necessary for the electric field induced negative ECE. This requires the temperature is close to the polarization peak temperature T_M . Around T_M , the free energy difference of the nonpolar and polar phases is quite small, which makes the metastable induced ferroelectric phase formation energetically favorable. As a result, the applied electric field triggers the phase transition and induces the significant enhancement in macroscopic polarizations, thus generates a negative ECE. Park et al. employed the GLD theory to explain the negative ECE observed in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ ferroelectric film.²⁰ Very recently, Pirc et al. performed the experimental and theoretical studies to understand the mechanism of the negative ECE in PbZrO_3 AFE ceramics based on the Kittel model.³⁰ Although more in depth atomistic simulation may help us more about the mechanism of negative ECE (for instance, the phase competition between AFE and FE, and oxygen octahedra rotation proposed in detail using the first principles simulations of ref.⁵⁴ might be the underlying mechanism in our studied material system), the

phenomenological Landau-type model could give us a quantitative explanation to the negative ECEs in antiferroelectrics and ferroelectrics. Our work here will also promote the practical applications of ECE in AFE bulk ceramics for room-temperature mid- and large-scale solid-state refrigerators.

4. CONCLUSIONS

In summary, giant negative EC responses in PLZST antiferroelectric bulk ceramics were realized, and a large ΔT up to -11.5 K near room temperature was achieved. The ΔT peak shifted toward lower temperatures with increasing electric field, which may be useful to realize electric field tunable EC response. The Ginzburg-Landau-Devonshire phenomenology was employed to explain the outstanding negative ECE, and a significant increase in polarization with increasing temperature was obtained, which contributes to the giant negative EC response. Our study demonstrated the promising potential of antiferroelectric ceramics for room temperature solid-state cooling devices. Combining both negative and positive EC responses in one material system may open up a new solution for better EC materials with high cooling efficiency in next-generation practical solid-state refrigerators.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website.

The SEM images of the fracture surfaces, temperature dependence of dielectric properties, Raman spectra, P - E hysteresis loops, electrocaloric temperature change, leakage current characterization, the comparison between electrocaloric temperature change and electrocaloric strength for the PLZST ceramics, and schematic plot of the three-dimensional and one-dimensional AFE and FE phases (PDF).

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Notes

The authors declare no competing financial interest.

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Table 1. The ECE Values and Corresponding Conditions for Various Materials.

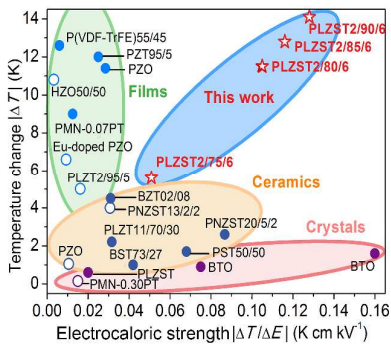
material ^{a)}		T	ΔT	ΔE	ΔS	ΔT/ΔE	ref
		[°C]	[K]	[kV cm ⁻¹]	[J kg ⁻¹ K ⁻¹]	[K cm kV ⁻¹]	
positive ECE	PbZr _{0.95} Ti _{0.05} O ₃ ^f	222	12	480	8	0.0250	9
	P(VDF-TrFE)55/45 ^f	80	12.6	2090	60	0.0060	10
	Pb _{0.8} Ba _{0.2} ZrO ₃ ^f	17	45.3	598	46.9	0.0758	11
	0.65PbMg _{1/3} Nb _{2/3} O ₃ -0.35PbTiO ₃ ^f	140	31	747	-	0.0415	23
	Pb _{0.97} La _{0.02} (Zr _{0.75} Sn _{0.18} Ti _{0.07})O ₃ ^f	5	53.8	900	63.9	0.0598	12
	PLZT8/65/35 ^f	45	40	1200	50	0.0333	24
	Ba(Zr _{0.20} Ti _{0.80})O ₃ ^{bc}	39	4.5	145	-	0.0310	25
	Pb _{0.99} Nb _{0.02} (Zr _{0.75} Sn _{0.20} Ti _{0.05}) _{0.98} O ₃ ^{bc}	161	2.6	30	-	0.0867	26
	BaTiO ₃ ^{bs}	139	1.6	10	-	0.1600	22
	Pb _{0.97} La _{0.02} (Zr _{0.66} Sn _{0.23} Ti _{0.11})O ₃ ^{bs}	160	0.6	30	-	0.0200	27
negative ECE	BaTiO ₃ ^{bs}	129	0.9	12	-	0.0750	28
	PMN-0.30PT ^{bs}	80	-0.15	10	-	0.0150	13
	Pb _{0.97} La _{0.02} (Zr _{0.66} Sn _{0.23} Ti _{0.11})O ₃ ^{bs}	125	-0.14	30	-	0.0047	16
	(Pb _{0.88} Sr _{0.08})[Nb _{0.08} (Zr _{0.53} Ti _{0.47}) _{0.92}]O ₃ ^b	80	-0.38	15	-	0.0253	29

^c

PbZrO ₃ ^{bc}	37	−1.05	100	-	0.0105	30
NBT ^{bc}	20	−1.6	70	-	0.0229	31
Pb _{0.99} Nb _{0.02} (Zr _{0.85} Sn _{0.13} Ti _{0.02}) _{0.98} O ₃ ^{bc}	82	−4.0	130	-	0.0307	32
(Pb _{0.97} La _{0.02})(Zr _{0.95} Ti _{0.05})O ₃ ^f	30	−5	308	-	0.0162	19
Hf _{0.5} Zr _{0.5} O ₂ ^f	175	−10.8	3260	−10.9	0.0033	20
4 mol% Eu-doped PbZrO ₃ ^f	130	−6.6	709	−5.42	0.0093	33
Pb _{0.97} La _{0.02} (Zr _{0.80} Sn _{0.14} Ti _{0.06})O ₃ ^{bc}	30	−5.5	110	−5.99	0.050	this work
Pb _{0.97} La _{0.02} (Zr _{0.80} Sn _{0.14} Ti _{0.06})O ₃ ^{bc}	30	−11.5	110	−12.5	0.105	this work
Pb _{0.97} La _{0.02} (Zr _{0.80} Sn _{0.14} Ti _{0.06})O ₃ ^{bc}	40	−12.9	110	−14.0	0.117	this work
Pb _{0.97} La _{0.02} (Zr _{0.80} Sn _{0.14} Ti _{0.06})O ₃ ^{bc}	50	−14.1	110	−13.5	0.128	this work

^{a)f} means thin films, *bs* indicates bulk single crystals, and *bc* refers to bulk ceramics.

TOC GRAPHIC



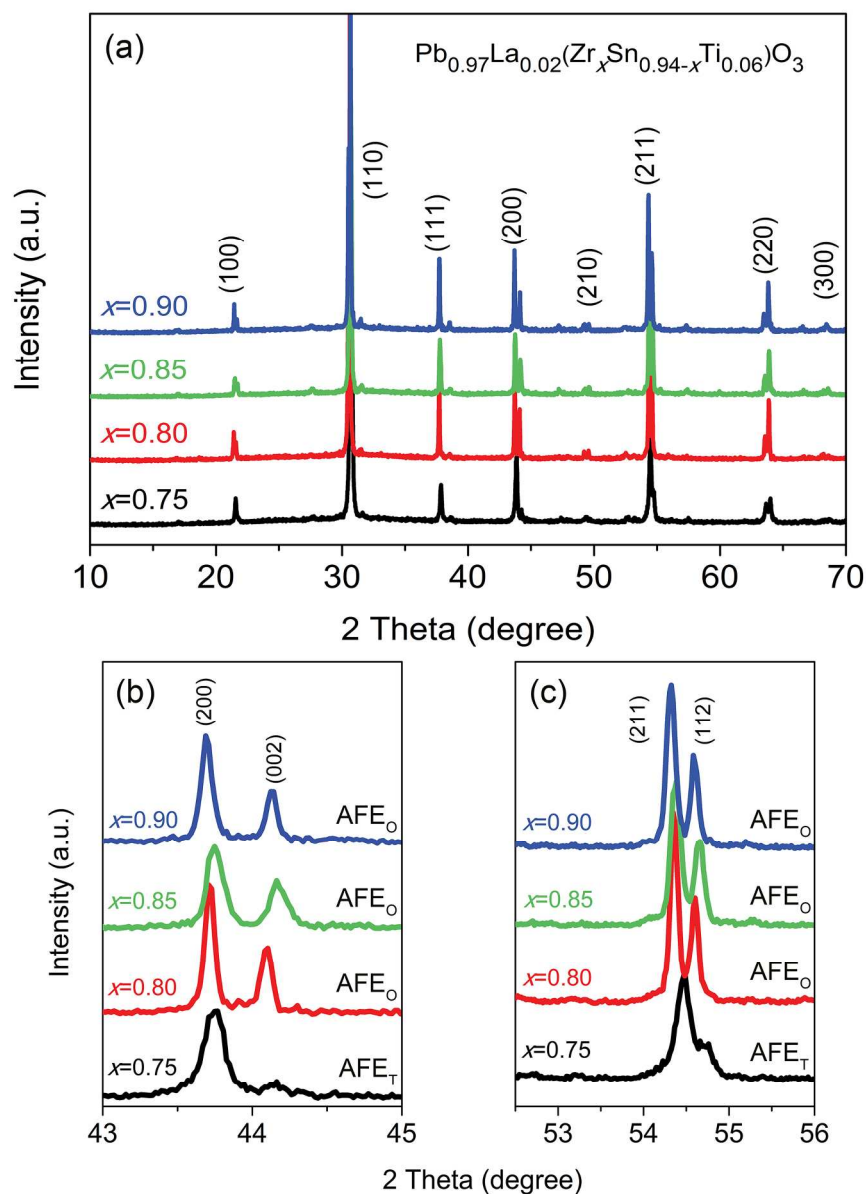


Figure 1. (a) X-ray diffraction patterns of the $\text{Pb}_{0.97}\text{La}_{0.02}(\text{Zr}_x\text{Sn}_{0.94-x}\text{Ti}_{0.06})\text{O}_3$ ceramics at room temperature. The fine scanning (b) (200)-diffraction peaks and (c) (211)-diffraction peaks.

170x232mm (300 x 300 DPI)

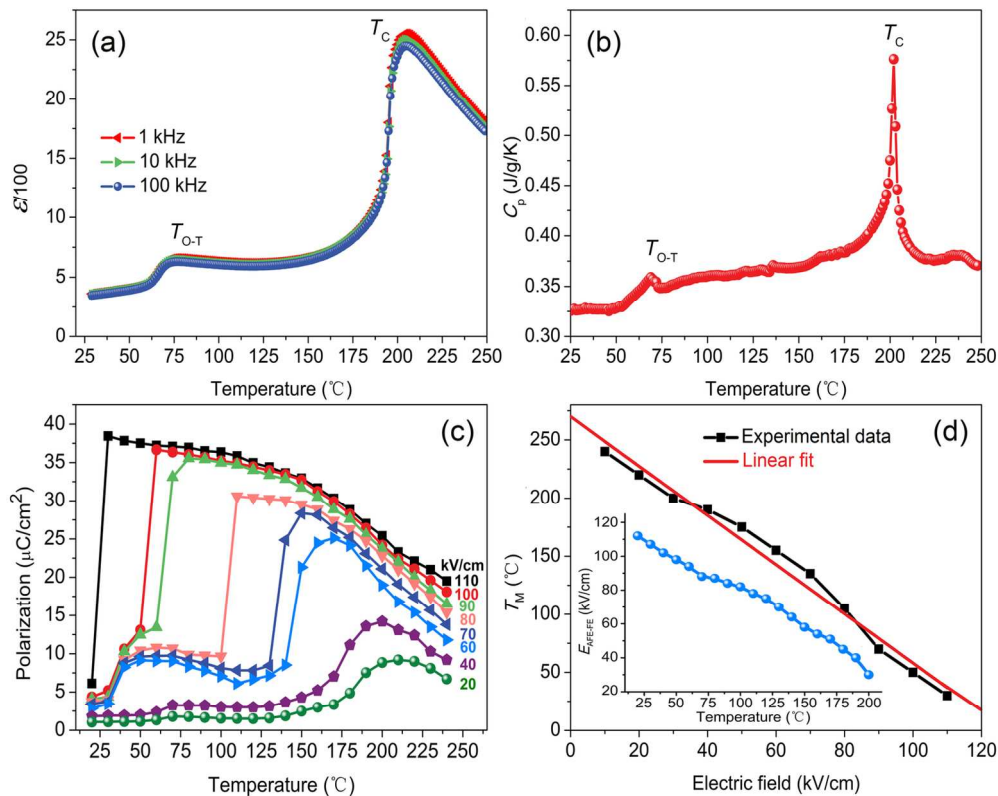


Figure 2. Temperature dependence of physical properties of PLZST2/80/06 ceramics: (a) dielectric permittivity (ϵ'); (b) specific heat (C_p); (c) Polarization. (d) The polarization peak temperature (T_M) as a function of electric field. The red solid line means a linear fit to the experimental data. The inset of (d) shows the temperature-dependent AFE-to-FE switching field (EAFE-FE).

137x111mm (300 x 300 DPI)

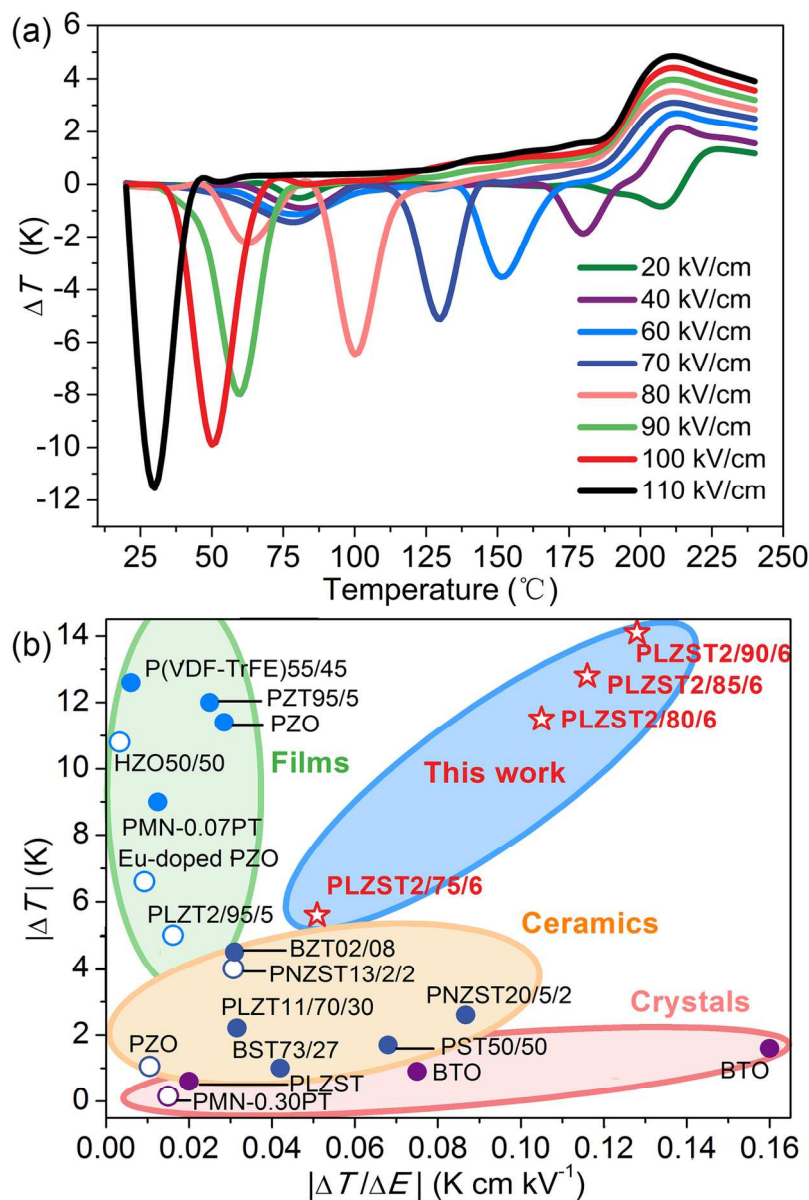


Figure 3. (a) Temperature change (ΔT) as a function of temperature at selected applied fields. (b) The magnitude of $|\Delta T|$ and EC strength $|\Delta T/\Delta E|$ for different materials. The open and filled symbols represent the negative and positive ECEs, respectively.

126x187mm (300 x 300 DPI)

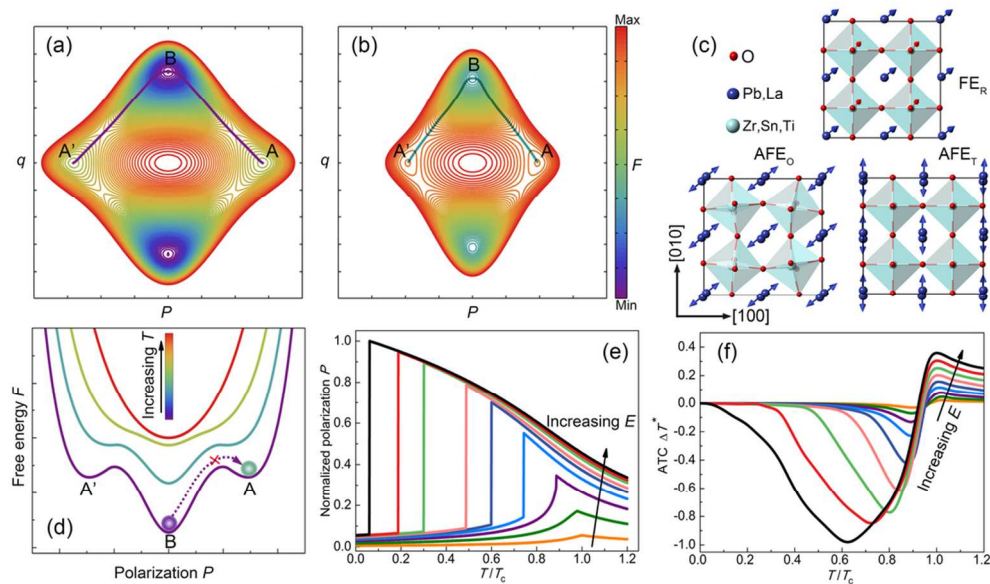


Figure 4. Contour plots of the free energy (F) at (a) $T < T_M$ and (b) $T > T_M$. (c) Relevant structures of $Pb_{0.97}La_{0.02}(Zr_xSn_{0.94-x}Ti_{0.06})O_3$: The AFE_O phase, schematically represented by the anti-polar distortions of A-site ions along $[110]$, and the oxygen octahedral tilt, the AFE_T phase, schematically characterized by the anti-polar distortions of A-site ions along $[010]$ and no oxygen octahedral tilts and the electric field induced FER phase, characterized by polar distortions along $[111]$. (d) Free energy–polarization (F – P) curves of antiferroelectrics for a negative ECE. The calculated results for (e) normalized polarization–temperature and (f) adiabatic temperature change $ATC (\Delta T^*)$ at selected electric fields.

99x58mm (300 x 300 DPI)

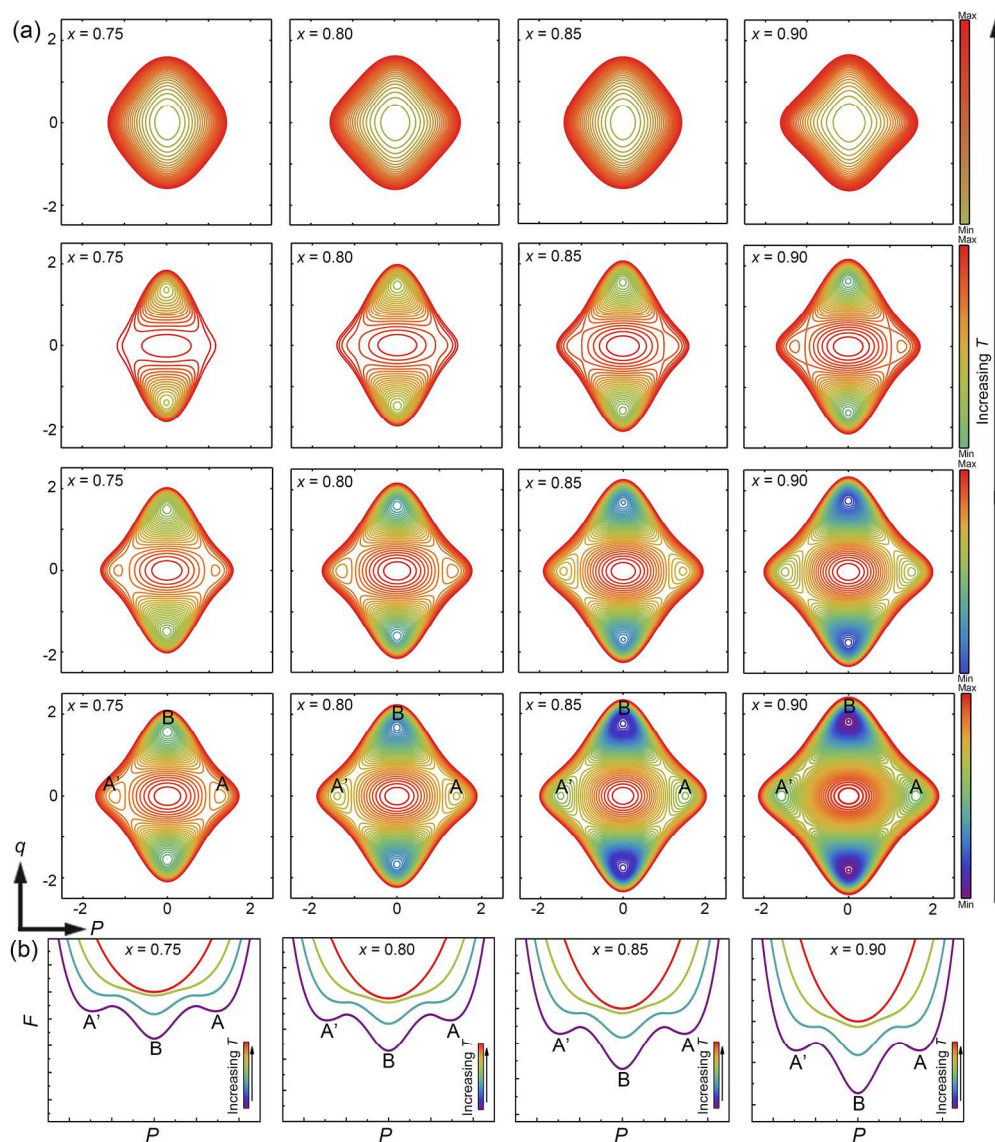


Figure 5. Ginzburg-Landau-Devonshire free energy phenomenology. (a) Contour plots of the free energy (F) with increasing composition (x) (left to right) and temperature (T) (bottom to top), described by eq 7 in Methods. The minimum free energy paths versus polarization (F - P) curves are contours plotted in (b).

180x206mm (300 x 300 DPI)